Exercise 5 :

. The ACF decreases very slowly and that assumes $\{Y_t\}$ is not stationary.

. under white noise assumption of $\{r_k\}$, we have

$$\sigma_r = SE(r_k) = \frac{1}{\sqrt{100}} = 0,1$$

using the $\pm 2\sigma_r$ 95% confidence interval yields that only r_1 is significative because

$$\forall\ 2 \leq k \leq 6,\ |r_k| < 2\sigma_r = 0,2$$

Conclusion the time serie is an ARIMA(0,1,1)